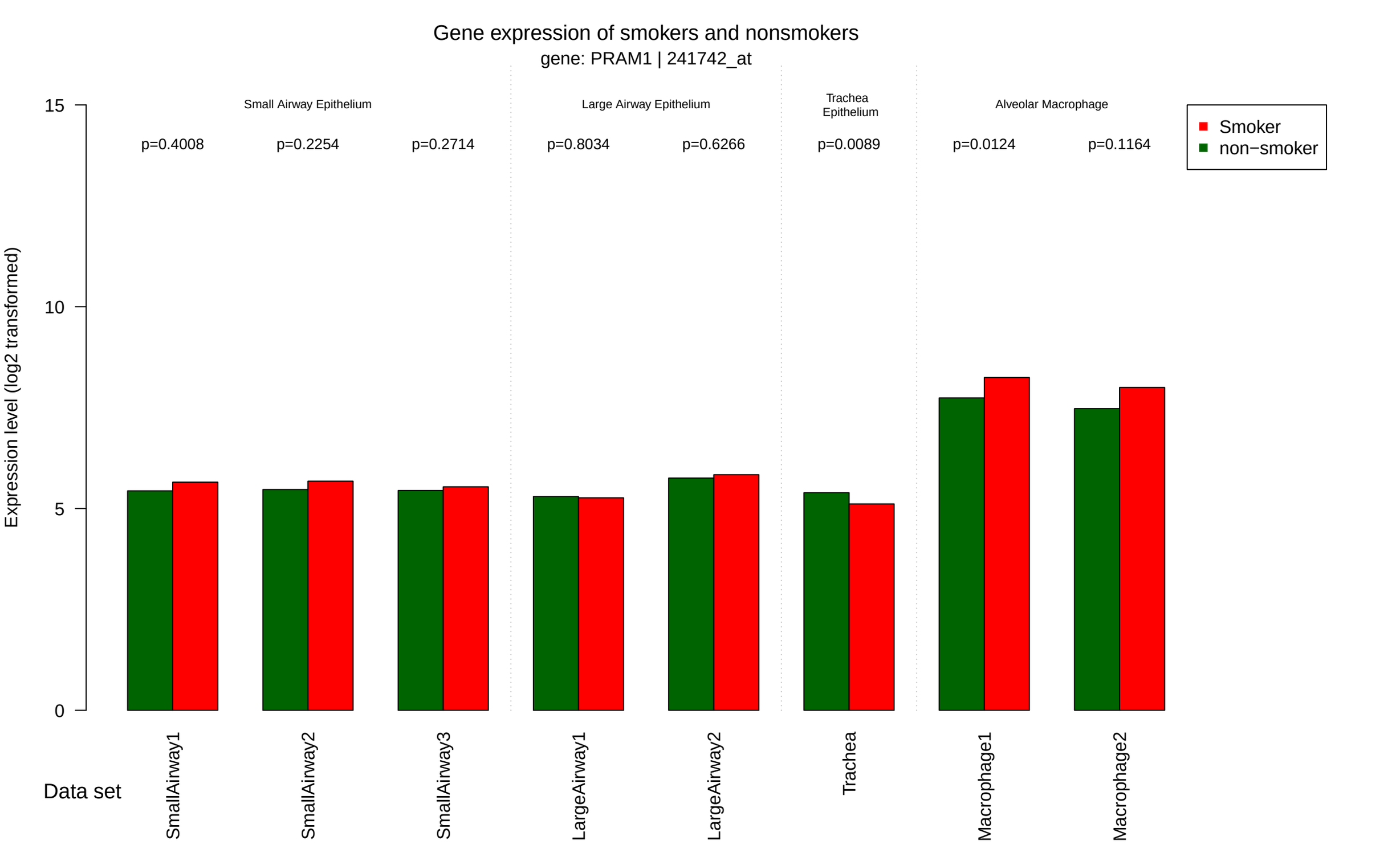
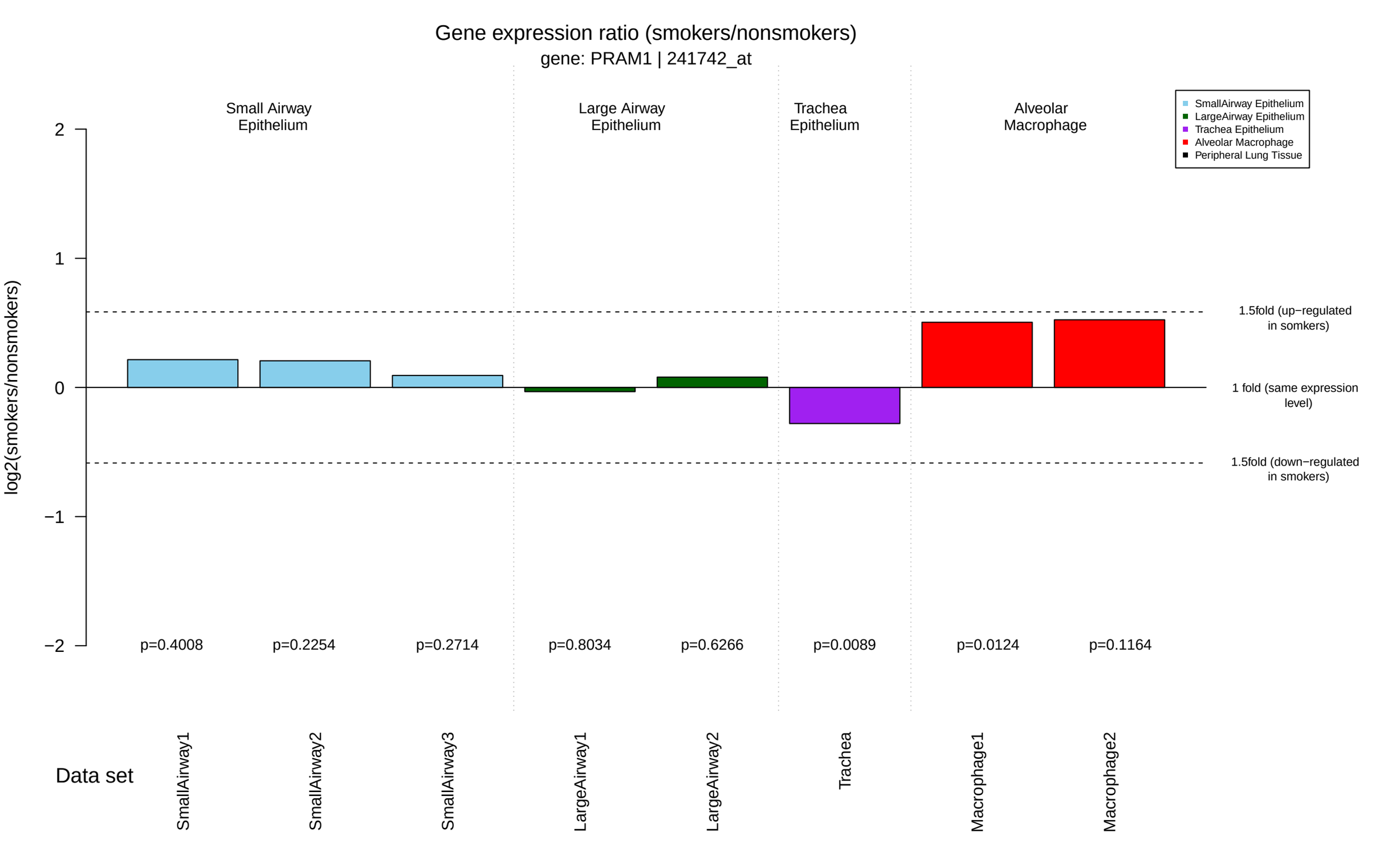
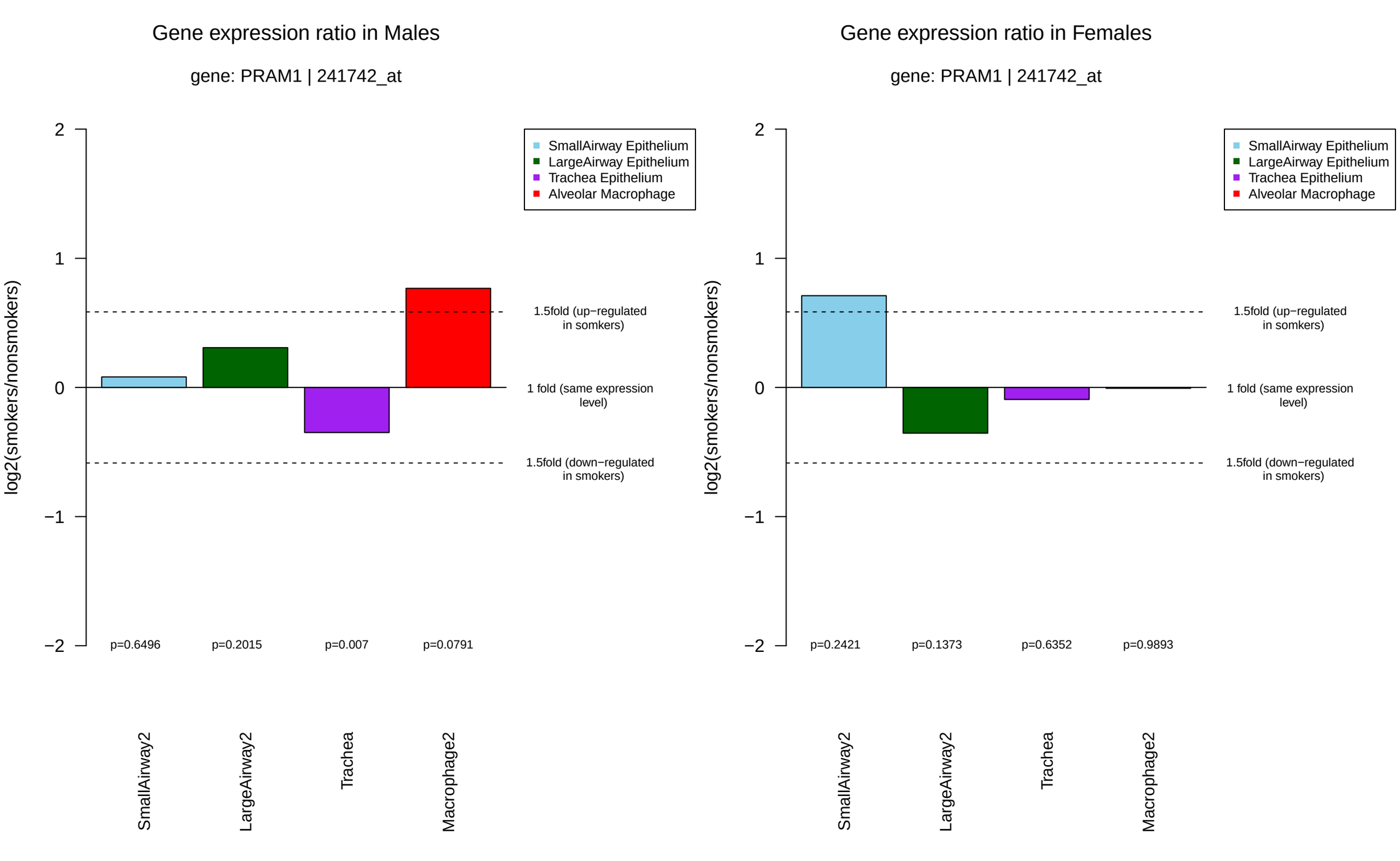
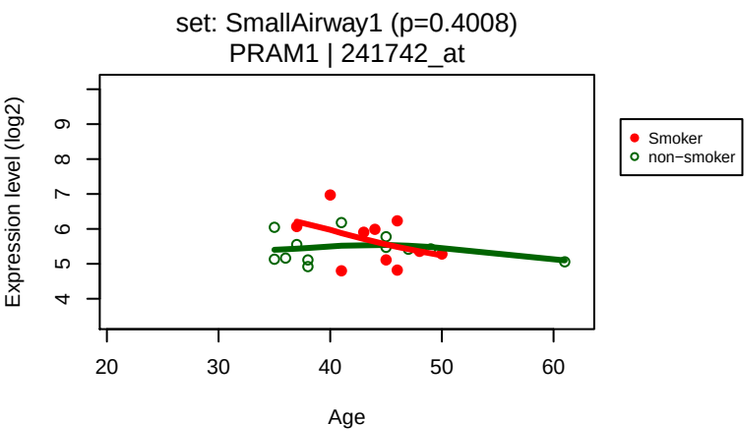
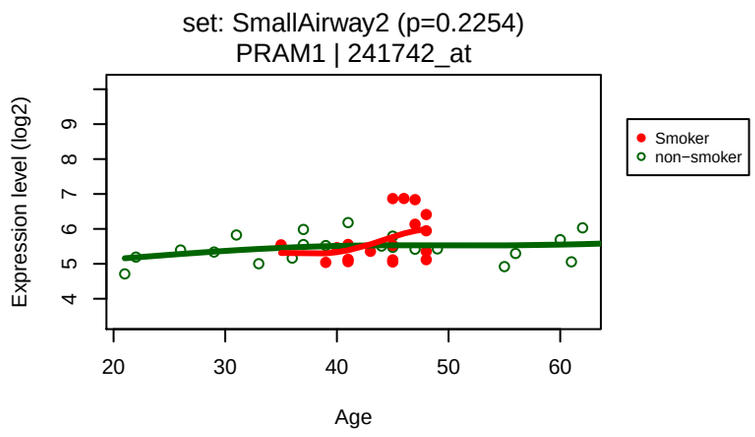
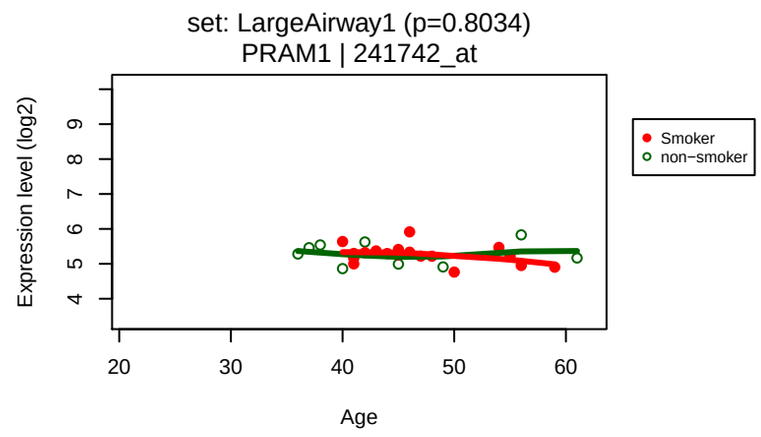

Gene expression of smokers and nonsmokers
gene: PRAM1 | 241742_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4008
p=0.2254
p=0.2714
p=0.8034
p=0.6266
p=0.0089
p=0.0124
p=0.1164
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

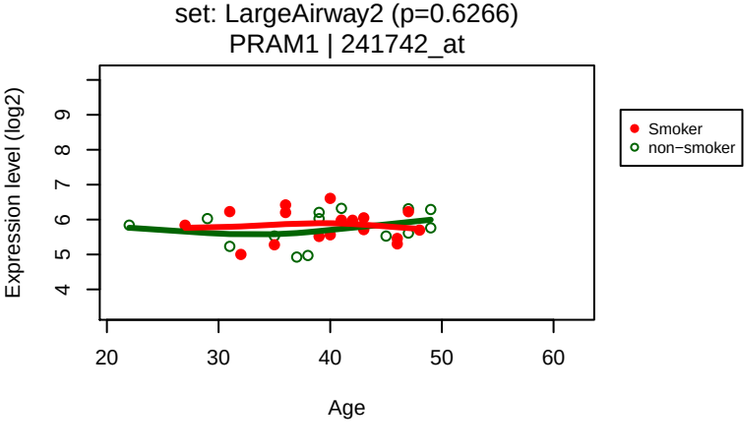

Gene expression ratio (smokers/nonsmokers)
gene: PRAM1 | 241742_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4008
p=0.2254
p=0.2714
p=0.8034
p=0.6266
p=0.0089
p=0.0124
p=0.1164
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

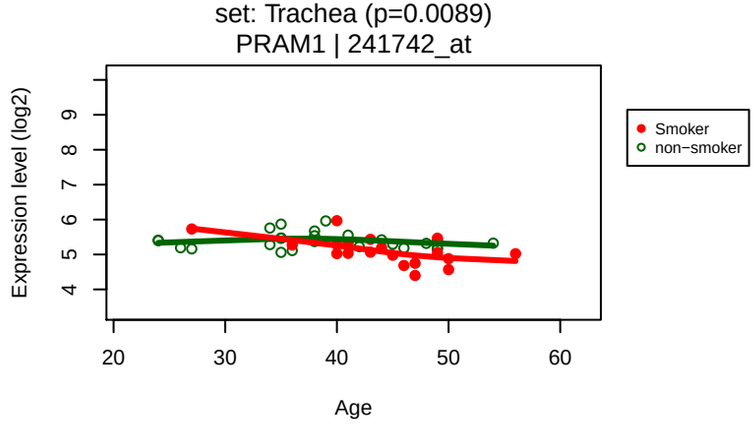

Gene expression ratio in Males
gene: PRAM1 | 241742_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6496
p=0.2015
p=0.007
p=0.0791
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRAM1 | 241742_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2421
p=0.1373
p=0.6352
p=0.9893
SmallAirway2
LargeAirway2
Trachea
Macrophage2

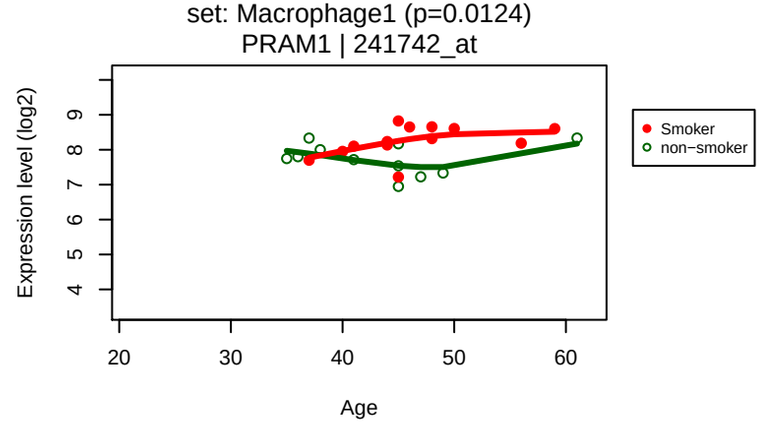

set: SmallAirway1 (p=0.4008)
PRAM1 | 241742_at
Expression level (log2)
Age
Smoker
non−smoker

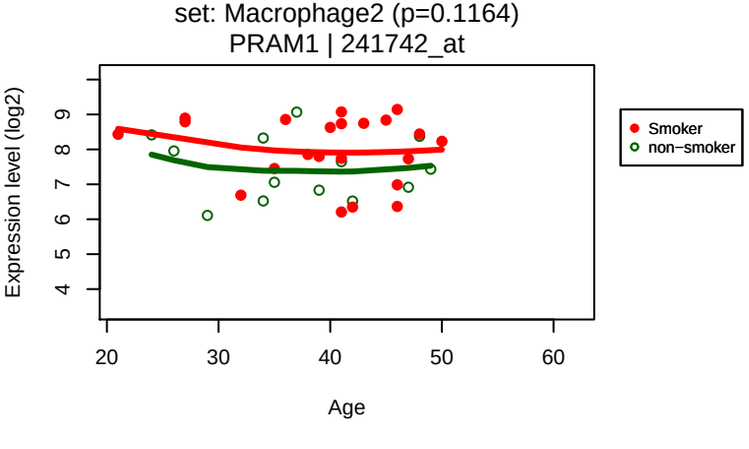

set: SmallAirway2 (p=0.2254)
PRAM1 | 241742_at
Expression level (log2)
Age
Smoker
non−smoker

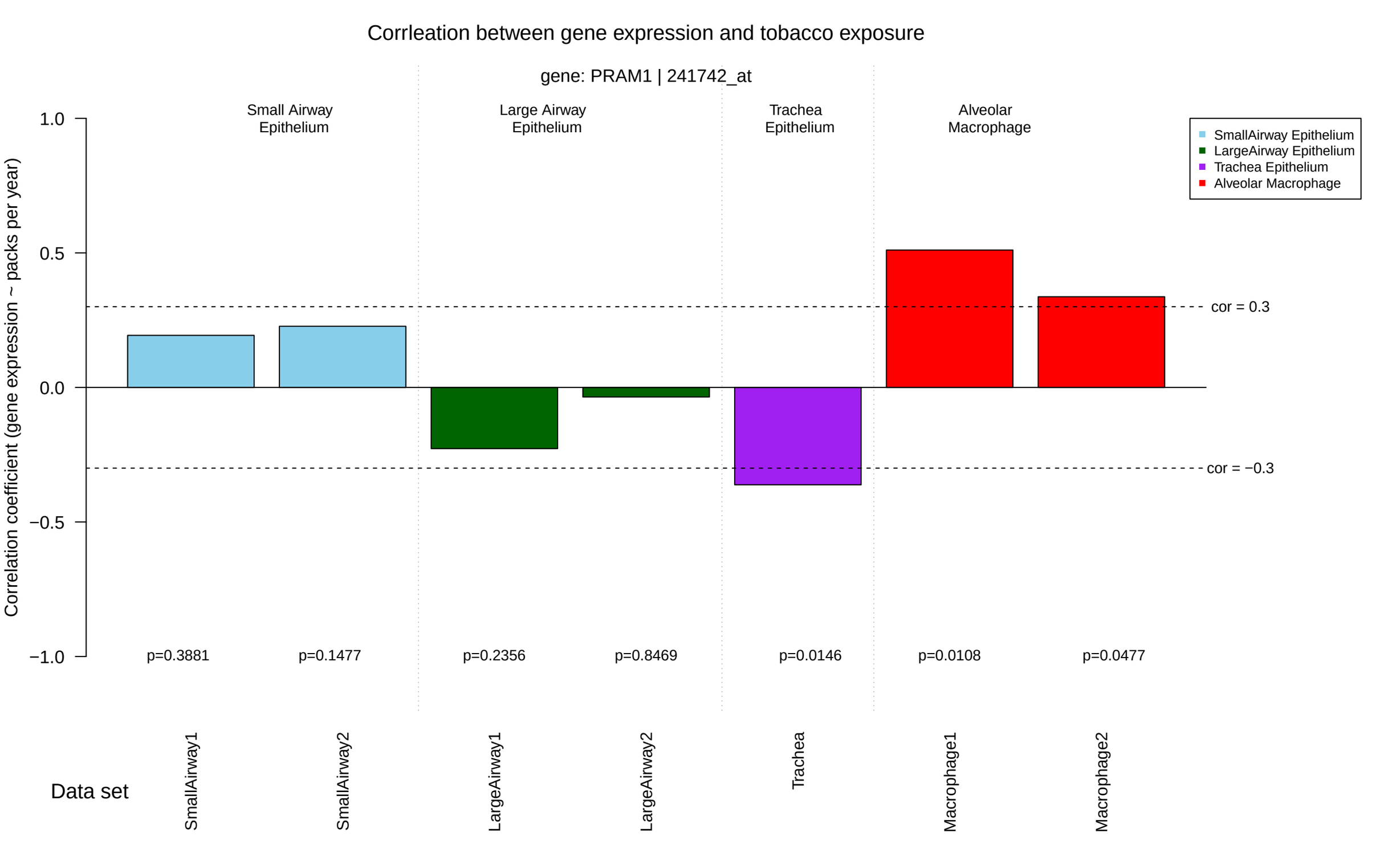

set: LargeAirway1 (p=0.8034)
PRAM1 | 241742_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6266)
PRAM1 | 241742_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0089)
PRAM1 | 241742_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0124)
PRAM1 | 241742_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1164)
PRAM1 | 241742_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRAM1 | 241742_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3881
p=0.1477
p=0.2356
p=0.8469
p=0.0146
p=0.0108
p=0.0477
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2